

Bhargav Srinivasan

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RESEARCH INTERESTS

I am interested in machine learning methods and theory for genomics, focusing on modeling and mechanistic interpretation of structural, epigenetic, and regulatory variation.

EDUCATION

University of Maryland

B.S. in Computer Science & B.S. in Biological Sciences

- Computer Science + Entomology Departmental Honors, Gemstone Honors Program

College Park, MD

Expected May 2027

GPA: 3.8/4

SELECT RESEARCH EXPERIENCES

Complex SV Identification — Kolmogorov Lab, National Cancer Institute

Student Intern

May 2026 – Present

Bethesda, MD

- Developing a deep learning method to identify and localize complex structural variation in cancer genomes.

Raw Signal Analysis Methods — Firtina Lab, UMD

Student Research Assistant

Oct 2025 – Present

College Park, MD

- Building methods for nanopore sequencing raw signal analysis: methylation detection, error correction, assembly.

Molecular Surveillance — AstraZeneca, Vaccines & Immune Therapies

Bioinformatics Intern

May 2025 – Aug 2025

Gaithersburg, MD

- Engineered and deployed a bacterial genome-processing pipeline supporting vaccine R&D surveillance efforts.
- Developed an unsupervised learning approach to identify non-typeable *Haemophilus influenzae* cell-type clusters.

US Department of Agriculture – Agricultural Research Service

Biological Science Aid

Apr 2024 – Jan 2025

Beltsville, MD

- Applied plant breeding methods to generate improved strawberry cultivars emphasizing disease resistance.
- Deployed computer vision models for real-time detection of diseased plants and stray animals with drones.

Insect Genome Bioinformatics Pipeline — Fritz Lab, UMD

Computational Biology Research Assistant

Feb 2024 – Present

College Park, MD

- Analyzing WGS short-read data to determine basis of pesticide resistance through copy number variation.
- Benchmarking short-read structural variation and copy number variation callers in collaboration with Molloy Lab.

Host-Pathogen Evolutionary Dynamics Model — Bruns Lab, UMD

Student Research Assistant

Jan 2024 – Present

College Park, MD

- Building ODE models to simulate host-pathogen coevolution to study temporal adaptation and allelic variation.

Fish Behavior Classification — Gemstone Program, UMD

Student Research Assistant, Mentor Liaison

Aug 2023 – Present

College Park, MD

- Developing a computer vision aided classification model to identify the social behaviors of Lake Malawi Cichlids.
- Team leader for 11 students and the main point of contact between PI Abhinav Shrivastava and team.

HONORS & AWARDS

1. ISMB HiTSeq Best Poster Award

RawMod: A Deep Learning Approach to Modification Agnostic Classification in Nanopore Sequencing

[ISMB: Intelligent Systems for Molecular Biology](#), Washington, D.C., USA. July 2026.

2. Entomological Society of Washington Student Showcase Best Poster Award

Correlations between Read Depth, Copy Number Variation, and Bt Resistance in Helicoverpa Zea

[Entomological Society of Washington Student Showcase](#), Washington, D.C., USA. May 2026.

3. Crop Science Society of America Poster Competition 2nd Prize

Correlations between Read Depth, Copy Number Variation, and Bt Resistance in Helicoverpa Zea

[PAG33: Plant and Animal Genome Conference](#), San Diego, CA, USA. January 2026.

4. University of Maryland Gemstone Library Awards Best Talk Award

Investigating the Link between the Ora4 Gene and Social Behaviors in African Cichlids through the Use of Computer Vision — awarded \$2,000 in research funding.

[University of Maryland Gemstone Library Awards](#), College Park, MD, USA. April 2025.

5. Recipient of the Honors College Research Grant, CS Professional Development Fund, GO Fund, and Ernest N. Cory Undergraduate Scholarship

Funds collectively totaling over \$5,000 in support of independent research and conference travel.

SELECT PUBLICATIONS

1. Simon Ambrozak, Ulysse McConnell, Bhargav Srinivasan, Burak Ozkan, Ernest Zhang, and Can Firtina, “*CRANE: Correcting Errors in Raw Nanopore Signals Using Hidden Markov Models*”, **arXiv**, 2026. [\[link\]](#)
2. Bhargav Srinivasan, Archana Venkataraman, and Srinivasa N. Raja, “*Artificial Intelligence and Pain Management: Cautiously Optimistic*”, [Editorial] **Pain Management**, Sep. 2024. [\[link\]](#)
3. Bhargav Srinivasan, Simon Ambrozak, and Can Firtina, “*RawMod: A Deep Learning Approach to Modification Agnostic Classification in Nanopore Sequencing*”, **in preparation**.
4. Bhargav Srinivasan, William Lamousin, Charles Phan, Jason Liu, Shubh Sharma, Vijay Jayasuriya, Ariel Bazan, Cade McGeehan, Ofure Osunbor, Tahir Haroon Pulkit Kumar, and Abhinav Shrivastava, “*FSHFormer: A Computer Vision Model for Automated Cichlid Behavior Annotation*”, **in preparation**.
5. Bhargav Srinivasan, Sam Hulse, and Emily Bruns, “*When should a pathogen strike? Navigating the landscape of birth pulses and mating season*”, **in preparation**.

SELECT PRESENTATIONS

1. **RawMod: A Deep Learning Approach to Modification Agnostic Classification in Nanopore Sequencing**
Poster Presentation, [ISMB: Intelligent Systems for Molecular Biology](#), Washington, D.C., USA. July 2026.
2. **A Deep Learning Approach to Identifying Complex Structural Variation in the Genome**
Invited Seminar Talk, [Cancer Data Science Laboratory](#), National Cancer Institute, Bethesda, MD, USA. July 2026.
Poster Presentation & Flash Talk, [NCI Summer School on Algorithmic Cancer Biology](#), Bethesda, MD, USA. July 2026.
3. **A Machine Learning Approach to Identify Unique Non-typeable Haemophilus influenzae (NTHi) Clusters Using Lipooligosaccharide (LOS) Biosynthetic Genes**
Poster Presentation, [ASM Microbe 2026](#), Washington, D.C., USA. June 2026.
4. **Unsupervised Learning for DNA Modification Profiles from Nanopore Sequencing Data**
Invited Talk, [Algorithmic Bioinformatics Seminar](#), University of Maryland, College Park, MD, USA. April 2026.
5. **Correlations between Read Depth, Copy Number Variation, and Bt Resistance in Helicoverpa Zea**
Poster Presentation, [PAG33: Plant and Animal Genome Conference](#), San Diego, CA, USA. January 2026.
Poster Presentation, [Entomological Society of Washington Student Showcase](#), Washington, D.C., USA. May 2026.
6. **Investigating color vision’s role in African Cichlid social behaviors using few-shot action recognition.**
Invited Talk, [Gemstone Do Good Showcase](#), College Park, MD, USA. November 2025.
7. **When Should a Pathogen Strike: Navigating the Landscape of Birth Pulses and Mating Season**
Poster Presentation, [Ecology and Evolution of Infectious Diseases North-East Meeting](#), University Park, PA, USA. November 2025.
8. **Developing a Molecular Surveillance Dashboard**
Invited Talk, [AstraZeneca Vaccines & Immune Therapies Intern Symposium](#), Gaithersburg, MD, USA. August 2025.
9. **Host Life History vs. Pathogen Optimization**
Invited Talk, [Quantitative Ecological & Evolutionary Dynamics Seminar](#), University of Maryland, College Park, MD, USA. April 2025.
10. **Investigating the Link between the Ora4 Gene and Social Behaviors in African Cichlids through the Use of Computer Vision**
Invited Talk, [University of Maryland Gemstone Library Awards](#), College Park, MD, USA. April 2025.

OPEN SOURCE TOOLS (MAIN CONTRIBUTOR)

All tools listed are actively in progress and under continuous development.

- **RawMod:** <https://github.com/bds062/Nanopore-Modification>
- **FSHFormer:** <https://github.com/jay11256/fsh-cluster>
- **Complex SV Detection:** <https://github.com/bds062/complex-SV>
- **Host–Pathogen Evolutionary Dynamics Model:**
<https://github.com/bds062/Host-Pathogen-Evolutionary-Dynamics-Model>